Molecular Toxicity Prediction via Fingerprints and Molecular Graph Networks: A Classical Machine Learning Baseline

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Code: https://github.com/mansingh-04/molecular-toxicity-prediction Demo: https://toxilab.man9876tr.workers.dev

***Abstract*—This work asks whether learned molecular graph representations outperform fixed fingerprint-based representations when both are evaluated under one data split and one protocol. Using the Tox21 benchmark (7,831 molecules and 12 binary assay endpoints, reduced to 7,708 after structure validation, salt stripping and duplicate merging), we establish a classical machine learning baseline. Logistic Regression, Random Forest and XGBoost were trained on Morgan fingerprints, RDKit descriptors and the two combined, with one model per endpoint. Missing labels were masked rather than treated as inactive, hyperparameters were selected on validation ROC-AUC, and the test set was evaluated once. Random Forest with combined features performed best, reaching a test ROC-AUC of 0.764 and average precision of 0.370; however, Logistic Regression achieved the highest ROC-AUC on 3 of the 12 endpoints, so no single model dominates. A Bemis–Murcko scaffold split was used, which places all acyclic molecules in the training partition, a limitation we report explicitly. Graph neural network results are not included in this phase and no graph results are claimed.**

***Index Terms*—molecular toxicity prediction, Tox21, Morgan fingerprints, random forest, XGBoost, scaffold split, class imbalance**

I. Introduction

Predicting whether a molecule is active in a biological assay is a well-established application of machine learning, and it is valuable because laboratory screening of every compound is slow and expensive. Two strategies exist for presenting a molecule to a model. The first converts the structure into a fixed numerical vector designed in advance, such as a Morgan fingerprint or a set of physicochemical descriptors. The second treats the molecule as a graph, with atoms as nodes and bonds as edges, and allows a neural network to learn its own representation during training.

The research question of this project is: *do learned molecular graph representations outperform fixed fingerprint-based representations when both are evaluated using the same data split and the same evaluation protocol?* The question is worth posing rather than assuming an answer. Published comparisons report that well-tuned descriptor-based models frequently remain competitive with graph models on molecular property tasks [4].

The project is divided into two phases. This paper reports Phase 1, which establishes a rigorous classical baseline. Phase 2 will implement a Graph Convolutional Network and measure it against that baseline under identical conditions. Our contributions are: (i) a cleaned, leakage-free Tox21 dataset with a frozen scaffold split shared by every model; (ii) an evaluation of three classical algorithms across three molecular representations with per-endpoint hyperparameter selection; and (iii) an honest account of the limitations, including an imbalance that makes accuracy uninformative and a scaffold split that excludes acyclic molecules from evaluation.

II. Related Work

The Tox21 challenge established a public benchmark for predicting activity across nuclear receptor and stress response pathways. Its multiple endpoints, incomplete labels and pronounced class imbalance make it a demanding test for molecular machine learning [1]. MoleculeNet subsequently collected Tox21 alongside other molecular datasets into a standard benchmark suite, and argued that scaffold-based splitting provides a more realistic estimate of generalisation than random splitting [2].

Classical molecular machine learning represents molecules using fixed descriptors or fingerprints. Extended-connectivity fingerprints encode local atomic environments as a fixed-length bit vector and remain among the most widely used representations in the field [3]. They are inexpensive to compute and pair naturally with established algorithms. Their limitation is that the representation is specified before training and therefore cannot adapt to the task.

Graph-based methods address this limitation by learning representations directly from molecular connectivity. Whether this yields a practical advantage is contested. Jiang *et al.* compared descriptor-based and graph-based models across eleven datasets and found that descriptor-based models, including gradient boosting, can match or exceed graph models on many endpoints [4]. That finding is the principal reason this project treats the comparison as an open question, and it motivates establishing a strong classical baseline before introducing a graph model.

III. Dataset and Preprocessing

The Tox21 dataset contains 12 binary classification endpoints covering nuclear receptor signalling (NR-AR, NR-AR-LBD, NR-AhR, NR-Aromatase, NR-ER, NR-ER-LBD, NR-PPAR-gamma) and cellular stress responses (SR-ARE, SR-ATAD5, SR-HSE, SR-MMP, SR-p53) [1]. Each label is active,

TABLE I: Cleaning funnel and resulting scaffold split

Cleaning step	Mols.	Partition	Mols.	Share
Raw dataset	7,831	Training	6,166	80%
Unsanitisable, removed	−8	Validation	771	10%
Duplicate rows merged	−115	Test	771	10%
Clean dataset	**7,708**	**Total**	**7,708**	100%

inactive, or not measured. Between 7% and 26% of labels are missing per endpoint, and only 3% to 16% of measured molecules are active.

Missing labels were retained as missing throughout. A label that was never measured is not evidence of inactivity, so each endpoint model was trained and evaluated using only those molecules with an observed label for that endpoint.

A. Cleaning

Table I summarises the effect of each cleaning step. Eight records were rejected by RDKit sanitisation. All eight are aluminium compounds whose SMILES write `[AlH3]` with three additional bonds, giving an explicit valence of six and exceeding the valence permitted for aluminium. They were removed rather than rewritten, since correcting them would require inventing structures absent from the source data.

Salts and mixtures were reduced to their largest fragment by heavy-atom count, affecting 244 records, and every structure was then written as a canonical SMILES string. This standardisation causes distinct source records to become identical: magnesium acetate, sodium acetate and potassium acetate all reduce to the acetate anion. Duplicate detection on canonical SMILES identified 72 such groups spanning 187 rows, which merged into 72 molecules and so removed 115 rows. Where merged records disagreed on an endpoint, the conflicting label was set to missing rather than resolved arbitrarily; this affected 47 labels. Merging is necessary because, without it, the same molecule could appear in both the training and test partitions under different salt forms, and reported performance would be optimistic.

B. Scaffold Split

A deterministic Bemis–Murcko scaffold split was used in place of a random split, following the recommendation of MoleculeNet [2]. Molecules were grouped by ring scaffold and whole groups assigned to a single partition, so that evaluation molecules contain ring systems absent from training. This is a harder and more realistic test of generalisation. Random seed 42 was used throughout, and no molecule, canonical SMILES or scaffold is shared between partitions.

A limitation follows from this procedure. Acyclic molecules have no ring scaffold and therefore share a single empty scaffold, which the algorithm assigns in its entirety to the training partition. All 1,753 acyclic molecules consequently sit in training, and the validation and test sets contain ring-containing molecules only. The results therefore say nothing about acyclic chemicals. The split was deliberately frozen in this state so that results from all team members remain directly comparable; changing it would require retraining every model.

IV. Exploratory Data Analysis

Label availability and class balance vary considerably across endpoints, as shown in Fig. 1. Missing rates range from 7.4% (NR-AR) to 25.8% (NR-Aromatase), and active rates from 2.9% (NR-PPAR-gamma) to 15.9% (SR-ARE).

Accuracy is not used as an evaluation metric in this work. With roughly one molecule in ten active, a model that predicts the majority class for every molecule already achieves approximately 89.5% accuracy. Accuracy therefore cannot distinguish a useful model from a trivial one on this data, and ROC-AUC and average precision are reported instead.

V. Molecular Representations

Three feature configurations were compared.

A. Morgan Fingerprints

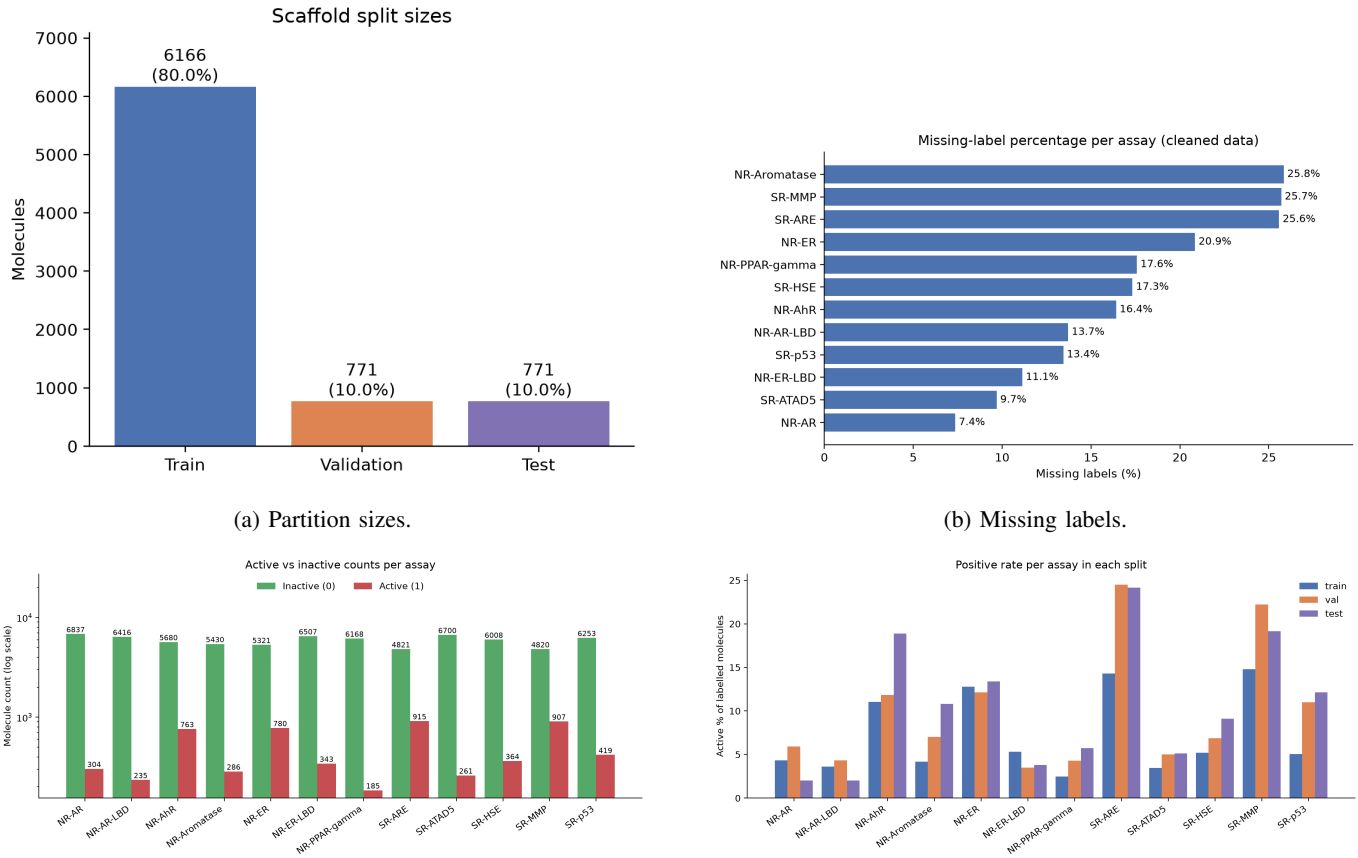
A Morgan fingerprint is a fixed-length bit vector derived from local atomic environments. Radius 2 and 2,048 bits were used, corresponding to the widely adopted ECFP4 setting [3]. Each heavy atom begins with an integer identifier derived from its element, degree, formal charge, attached hydrogens and ring membership. At each of the two iterations, every atom's identifier is recomputed from its own previous identifier together with the bond types and identifiers of its neighbours. After two iterations, an identifier summarises the substructure within two bonds of that atom. Every identifier generated is folded into the bit vector by taking its remainder modulo 2,048, and the corresponding bit is set to one.

Because far more substructures exist than available bit positions, two distinct environments can map to the same position. A set bit therefore indicates that at least one substructure hashing to that position is present, rather than identifying a unique fragment. These hash collisions are an accepted cost of obtaining a compact, fixed-length representation.

B. Molecular Descriptors and Combined Features

Twelve RDKit descriptors summarise global physicochemical and structural properties: molecular weight, lipophilicity (MolLogP), hydrogen-bond donors and acceptors, topological polar surface area, rotatable bonds, ring count, aromatic rings, heavy atoms, heteroatoms, fraction of sp^3 carbons, and formal charge. All are computed from the two-dimensional molecular graph, so no conformer generation is required. Median imputation and standardisation were fitted on the training partition only, ensuring that no evaluation data influences preprocessing.

The combined representation concatenates the 2,048 fingerprint bits with the twelve descriptors, giving 2,060 features. Fingerprint bits are passed through unscaled; only the descriptor columns are imputed and standardised.



(a) Partition sizes.

(b) Missing labels.

(c) Active vs inactive (log).

(d) Active rate per partition.

Fig. 1: Label availability and class balance across the 12 Tox21 endpoints.

VI. Models and Experimental Protocol

Three classical algorithms were selected to represent distinct model families rather than variations of a single idea. **Logistic Regression** serves as the linear baseline; it is interpretable, since each weight indicates how a fingerprint bit shifts the log-odds of activity, and it establishes the performance that a more complex model must exceed to justify its complexity. **Random Forest** is a bagging ensemble in which many deep trees are grown on bootstrap samples, with a random subset of features considered at each split, and their predictions averaged; this reduces variance and captures interactions between substructures that a linear model cannot. **XGBoost** is a boosting ensemble in which trees are built sequentially, each fitting the gradient of the loss remaining from its predecessors, reducing bias rather than variance. One binary classifier was trained per endpoint, giving 36 models per feature configuration.

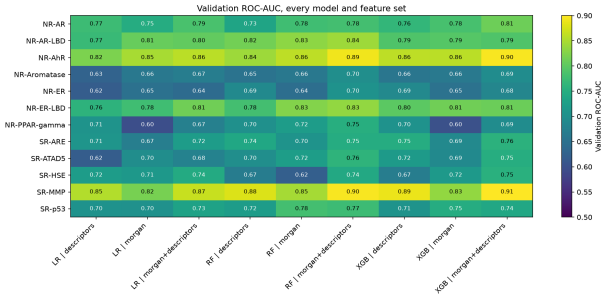
A. Class Imbalance

The positive class is rare for every endpoint, so an unweighted model can reduce its loss largely by predicting the negative class. Class weighting was therefore treated as a tunable hyperparameter rather than assumed beneficial, using `class_weight` for Logistic Regression, `balanced_subsample` for Random Forest and `scale_pos_weight` for XGBoost. For each endpoint the setting was selected on validation performance.

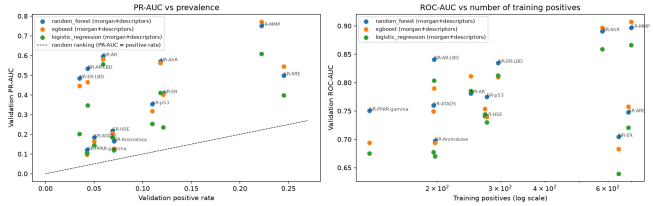
B. Hyperparameter Selection and Evaluation

For Logistic Regression the regularisation parameter was searched over $C \in \{0.001, 0.01, 0.1, 1, 10\}$ with class weight in {none, balanced}, using the `lbfgs` solver with a maximum of 5,000 iterations. Random Forest used 300 trees with `max_features` = `sqrt` and unlimited depth, searching minimum leaf size over $\{1, 3\}$ and class weight over {none, balanced_subsample}. XGBoost searched maximum depth over $\{3, 6\}$, learning rate over $\{0.05, 0.1\}$, number of estimators over $\{100, 300\}$ and `scale_pos_weight` over $\{1, \text{balanced}\}$, with subsample 0.8, column sample 0.5 and L2 regularisation 1.0.

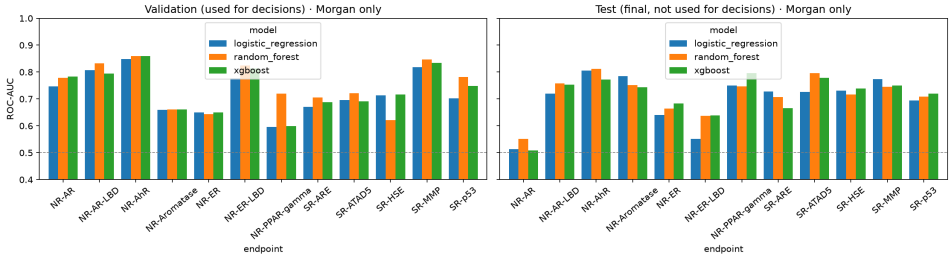
Validation ROC-AUC was the primary model-selection metric, with average precision as a tie-breaker. ROC-AUC measures the probability that a randomly chosen active molecule is ranked above a randomly chosen inactive one and is unaffected by class prevalence; average precision summarises the precision-recall curve and is more sensitive to performance on the rare positive class. All hyperparameter choices were fixed using validation data before any test prediction was made, so the test set was used exactly once.



(a) Validation ROC-AUC, all configurations.



(b) Imbalance against performance.



(c) ROC-AUC per endpoint.

Fig. 2: Model comparison across feature configurations and endpoints.

TABLE II: Validation and test performance, averaged over 12 endpoints

Model	Features	Val. AUC	Val. AP	Test AUC	Test AP
Log. Regression	Morgan	0.724	0.323	0.702	0.277
Log. Regression	Descriptors	0.723	0.222	0.666	0.219
Log. Regression	Combined	0.749	0.297	0.739	0.269
Random Forest	Morgan	0.750	0.382	0.716	0.317
Random Forest	Descriptors	0.743	0.333	0.736	0.301
Random Forest	**Combined**	**0.785**	**0.407**	**0.764**	**0.370**
XGBoost	Morgan	0.737	0.349	0.712	0.281
XGBoost	Descriptors	0.749	0.337	0.710	0.269
XGBoost	Combined	0.774	0.391	0.755	0.354

TABLE III: Test ROC-AUC per endpoint, Morgan fingerprints

Endpoint	Obs.	Pos.	LR	RF	XGB
NR-AR	691	14	0.514	**0.552**	0.509
NR-AR-LBD	598	12	0.719	**0.758**	0.754
NR-AhR	582	110	0.806	**0.811**	0.772
NR-Aromatase	490	53	**0.785**	0.751	0.744
NR-ER	537	72	0.641	0.664	**0.683**
NR-ER-LBD	631	24	0.551	0.637	**0.638**
NR-PPAR-gamma	559	32	0.750	0.747	**0.797**
SR-ARE	497	120	**0.728**	0.708	0.666
SR-ATAD5	645	33	0.725	**0.797**	0.778
SR-HSE	605	55	0.731	0.716	**0.739**
SR-MMP	465	89	**0.774**	0.746	0.751
SR-p53	618	75	0.694	0.709	**0.721**

VII. Results

Table II reports performance averaged over the 12 endpoints. Random Forest with combined features was strongest, achieving a validation ROC-AUC of 0.785 and a test ROC-AUC of 0.764, together with the highest test average precision of 0.370. XGBoost with combined features followed at 0.755 and 0.354. For all three algorithms the combined representation outperformed either fingerprints or descriptors alone, indicating that local substructure patterns and global physicochemical properties carry complementary information. Fig. 2 summarises these comparisons visually.

Table III gives per-endpoint results. Performance varies considerably across biological targets. Discrimination is strongest for NR-AhR (0.811) and SR-ATAD5 (0.797), and weakest for NR-AR, which has only 14 positive examples in the test partition; with so few positives that estimate is unstable and should not be over-interpreted. The same caution applies to NR-AR-LBD, which has 12.

Notably, Logistic Regression achieves the highest ROC-AUC on 3 of the 12 endpoints: NR-Aromatase, SR-ARE and SR-MMP. The ensembles therefore do not outperform the linear baseline everywhere, which is a useful reminder that additional model complexity is not uniformly beneficial.

A. Reproducibility

All experiments used random seed 42 with Python 3.13, scikit-learn 1.6.1, XGBoost 3.4.1 and RDKit 2026.09. Re-running the preprocessing notebook on the raw Tox21 file reproduces the cleaned dataset, the three split files and the preprocessing report byte for byte, which was verified while preparing this paper. Re-running the model notebooks reproduces the Logistic Regression results exactly; Random Forest and XGBoost differ by at most 1×10^{-5} ROC-AUC because parallel tree construction depends on the CPU and library build, which changes no reported value at three decimal places.

TABLE IV: Division of work across the team

Member	Responsibility and deliverables
Khushi	Data loading and inspection, SMILES validation and standardisation, duplicate and conflicting-label audit, exploratory data analysis. Notebook 01; cleaned dataset; EDA figures.
Riya Yadav	Data cleaning and exploratory analysis; Bemis–Murcko scaffold split and leakage checks; project documentation. Notebook 01; split files; this report.
Manpreet Singh	Morgan fingerprint and molecular descriptor generation; shared feature interfaces; web application and deployment. Feature code in notebook 02; web application.
Avneet Singh	Model selection and training for Logistic Regression and Random Forest; hyperparameter search on validation data. Notebooks 02 and 03; trained models.
Satvik Prasad	Model selection and training for XGBoost; endpoint-level evaluation, model comparison and class-imbalance analysis. Notebook 03; metrics and comparison tables.

All trained models, metrics and per-molecule predictions are published in the project repository.

VIII. Contribution of Team Members

Table IV sets out how the work was divided. All members contributed to interpreting the results and to preparing this paper.

IX. Conclusion and Limitations

This paper establishes a rigorous classical baseline for molecular activity prediction on Tox21. Combining Morgan fingerprints with physicochemical descriptors improved performance for all three algorithms relative to either representation alone, and Random Forest with combined features was strongest overall at a test ROC-AUC of 0.764 and average precision of 0.370. No single model is best everywhere: Logistic Regression led on three endpoints while the ensembles led on the remaining nine.

Several limitations qualify these results. They derive from a single split and a single random seed without confidence intervals, so small differences between models may not be meaningful. NR-AR and NR-AR-LBD have only 14 and 12 test actives respectively, making those estimates unstable. Acyclic molecules are confined to the training partition, so the results do not characterise them. Only classical models are covered, and no graph results are claimed. Finally, performance on this benchmark does not establish real-world human toxicity.

Phase 2 will implement a Graph Convolutional Network with a masked loss that excludes missing labels, trained on the same scaffold split and evaluated with the same metrics, so that the comparison with this baseline is fair. A hybrid model combining fingerprint features with graph-learned representations is a further possibility, to be pursued only if it improves on both components individually.

Responsible Use

The models presented here predict activity in specific *in-vitro* Tox21 assays. They are research outputs, not experimental measurements, safety assessments or statements about human toxicity, and must not be used for regulatory, clinical or medical decisions.

References

[1] R. Huang, M. Xia, *et al.*, "Tox21Challenge to build predictive models of nuclear receptor and stress response pathways as mediated by exposure to environmental chemicals and drugs," *Frontiers in Environmental Science*, vol. 3, art. 85, 2016, doi: 10.3389/fenvs.2015.00085.

[2] Z. Wu, B. Ramsundar, *et al.*, "MoleculeNet: a benchmark for molecular machine learning," *Chemical Science*, vol. 9, no. 2, pp. 513–530, 2018, doi: 10.1039/C7SC02664A.

[3] D. Rogers and M. Hahn, "Extended-connectivity fingerprints," *Journal of Chemical Information and Modeling*, vol. 50, no. 5, pp. 742–754, 2010, doi: 10.1021/ci100050t.

[4] D. Jiang, *et al.*, "Could graph neural networks learn better molecular representation for drug discovery? A comparison study of descriptor-based and graph-based models," *Journal of Cheminformatics*, vol. 13, art. 12, 2021, doi: 10.1186/s13321-020-00479-8.